

P5 Planning, analysis and evaluation

Practical · 2026 examination · Concept guide and original exam-style practice

Understand the concept

A plan identifies reaction, independent and dependent variables, controls and a measurable endpoint. Use a graph to distinguish orders or calculate an enthalpy with $q=mc\Delta T$, then evaluate heat losses and significant uncertainty. Safety controls should address actual reagents, not an unconnected generic warning.

Worked example: follow the reasoning

For a rate trial, measure time for a fixed turbidity threshold while changing concentration but holding volume and temperature constant. Use reciprocal time only as a proportional rate estimate when endpoint amount is constant.

Quick check · multiple choice

A For a fair rate comparison, what must be controlled?

[1]

- A Two variables changed together
- B Initial-rate units omitted
- C Temperature
- D Graph slope hidden

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Outline a fair comparison of reaction rate at two acid concentrations using gas volume.

[3]

Answers and explanation

Quick check: C. Temperature

Question 1 [3 marks]

Keep acid volume, solid mass/area and temperature constant [1]; change acid concentration [1]; measure initial gradient of gas-volume against time with repeat trials [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.